

Xuanpu Zhang

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PROFESSIONAL SUMMARY

AI systems engineer and NUS Master of Computing student focused on multi-agent orchestration, LLM harnesses, and AI-enabled productivity. Founded and independently built Skip Lecture, an AI study platform serving 120 users, including 10 paid users; redesigned an agent harness around observable, interruptible workflows. Combine Python/TypeScript product engineering with grounded retrieval, LLM evaluation, and model fine-tuning. Led a three-person robotics team and university teaching initiatives, bringing hands-on technical ownership, team coordination, and experience turning AI capabilities into usable products.

TECHNICAL COMPETENCY

Agent Systems: Multi-agent orchestration, MCP, Tool use, Persistent conversation state, Scoped workspaces, Human-in-the-loop control.

LLM Engineering: Grounded retrieval, Structured OCR, Citation mapping, Prompt/context engineering, Model routing, LLM evaluation.

Software Engineering: Python, TypeScript, Node.js, React, FastAPI, PostgreSQL, Redis, Celery.

Systems and Delivery: Docker, AWS S3, Linux, Git, GitHub Actions.

Languages: Chinese (Native), English (Business), French (Beginner).

EDUCATION

National University of Singapore

Master of Computing in Artificial Intelligence

Singapore

Aug. 2026 - Jan. 2028

McGill University

Bachelor of Science in Computer Science - Artificial Intelligence; Minor in Mathematics

Montreal, Canada

Sept. 2022 - May 2026

- Relevant coursework: Machine Learning, Natural Language Processing, Artificial Intelligence, Reinforcement Learning, Algorithms and Data Structures, Software Design, Operating Systems, Database Systems, Convex Optimization.

PROFESSIONAL EXPERIENCE

Skip Lecture - AI Study Platform

Jul. 2025 - Present

Role: Founder; Sole full-stack AI product engineer

- Shipped and grew a self-developed AI study platform to 120 users, including 10 paid users, owning product design, engineering, deployment, and operations end to end.
- Designed a token-gated multi-model pipeline that converts PDF pages into structured JSON/Markdown with Gemini 2.5 Flash before GPT5.4 synthesis, with checkpoint/resume and corrupt-output recovery.
- Built hybrid retrieval and stable file-hash-plus-page citations across notes, slide OCR, video frames, transcripts, and quoted context to keep generated answers tied to source material.
- Architected an asynchronous FastAPI/SQLModel/PostgreSQL backend with Celery, Redis, and S3, including active-task recovery, row-locked token accounting, and Stripe-idempotent billing.
- Delivered the React/TypeScript product experience for multi-source uploads, rich-text editing, citation navigation, sharing, bilingual flows, and note-grounded chat.

China Mobile

Jun. 2025 - Aug. 2025

AI Training Engineer Intern

Zhengzhou, China

- Raised Qwen3-8B Chinese-Thai translation BLEU-style scores from the teens to the twenties by curating domain data and applying LoRA fine-tuning with LLaMAFactory.
- Brought the fine-tuned model close to the translation quality of the publicly available full Qwen3 model in the project's translation tests.
- Curated Chinese-Thai news translation pairs and configured the fine-tuning and evaluation workflow to compare baseline and adapted model outputs.
- Diagnosed and corrected Thai Buddhist-era versus Gregorian year translation errors, addressing a recurring domain-specific localization failure.

ISS Tech

Jun. 2024 - Aug. 2024

Software Developer Intern

Zhengzhou, China

- Doubled A100 real-time compression capacity from 12 to 24 concurrent video streams by tuning FFmpeg encoder parameters for a government video-surveillance aggregation platform under preparation.
- Tuned compression settings to balance high compression ratios with consistent video quality, supporting the platform's requirements for high-quality, reviewable footage.
- Enabled simultaneous multi-stream compression with smooth real-time playback and preview for monitoring workflows.
- Developed within a GPU-assisted streaming stack using C/C++, Linux, Bash, and Android Studio, balancing compression throughput, playback responsiveness, and visual quality.

TECHNICAL EXPERIENCE

Remote Lab / huiyuanClaw - Agent Orchestration Harness

Feb. 2026 - Present

Role: Agent orchestration and developer-tools builder

- Made concurrent agent execution observable and interruptible by exposing each delegated run as a separate conversation, enabling users to monitor progress, intervene directly, or terminate abnormal work.
- Redesigned Ninglo/remotelab into a browser-based orchestration harness with MCP-spawned conversations, persistent state, and controllable concurrent workflows.
- Scoped agents to project/folder workspaces with smaller skill and instruction surfaces, reducing unnecessary context use while improving execution control.
- Extended the Node.js/WebSocket/MCP/JSONL runtime with persistent sessions, task dependencies, browser control, and reusable team workflows.
- Integrated Git-aware diffs, staging, commits, branches, pull/push operations, plus Cloudflare Tunnel, ttyd, and dtach access for auditable remote delivery.

Embodied Intelligence - LLM-Powered Human-Robot Interaction

Jun. 2025 - Aug. 2025

Role: Team lead; LLM/MCP system owner

- Led a 3-person Raspberry Pi robot project to a voice-to-action loop that initiated tasks within 3 seconds in internal demos, while owning the LLM/MCP and ROS integration path.
- Defined interface boundaries across STT, LLM/MCP routing, function-level commands, ROS execution, and TTS feedback so separately owned components operated as one pipeline.
- Deployed and tuned a low-resource language model on Raspberry Pi, validating function-call behavior in simulation before connecting outputs to physical execution.

SELECTED PROJECTS RELATED TO JOB

COMP 550 NLP - Long-Context LLM Evaluation

Mar. 2026 - Apr. 2026

Role: Experiment design lead; evaluation implementation owner

- Built a four-condition long-context evaluation that showed exact evidence-chain recovery falling to zero in the longest setting, exposing a clear GPT-4o reliability limit.
- Proposed and led the lecture-slide study, defining evidence-grounded logical-chain tasks with exact page-ID citation requirements.
- Implemented structured JSON/POML artifacts and sequence/set metrics including Needleman-Wunsch alignment, LCS recall, and precision/recall/F1.

COMP 579 - Offline RL Reproduction

Mar. 2026 - Apr. 2026

Role: Experiment environment and server owner

- Benchmarked 4 offline-RL approaches across 3 dataset regimes and 2 environment families, showing fixed-alpha CQL gains on medium-expert MuJoCo while CQL-Lagrange remained unstable.
- Owned server and experiment setup for the PyTorch reproduction, coordinating parallel runs across D4RL MuJoCo and MiniGrid settings.

MATH 463 - Image Recovery Optimization

Mar. 2026 - Apr. 2026

Role: Validation, reporting, and team-organization lead

- Converted 2 optimization workflows into reproducible image-recovery experiments with objective tracking, convergence curves, and visual comparisons.
- Coordinated team execution and owned validation/reporting for constrained deblurring and denoising experiments.

OTHER EXPERIENCE

McGill University - COMP 202 Team Mentor (Undergrad TA)

Sept. 2023 - May 2026

Role: Undergraduate teaching assistant / Team Mentor

- Led weekly office hours serving 10-20 student visits per session and approximately 200 visits per semester, guiding Python debugging, assignment reasoning, and concept clarification.
- Co-designed assignment problems in three-person teams, often owning the main problem content; reviewed approximately 10 assignments over three years.
- Led the structure and template for a 24-document Python review archive and organized roughly half of the materials for future student preparation.

WuTong Cup Data Security - Anomaly Detection

2023

Role: Team lead

- Won First Prize in the Shanghai Municipality Data Safety Category by leading a 3-member team to deliver roughly 99% task coverage with about 13-second processing time.
- Built speed-oriented Python/Pandas data workflows while evaluating coverage and runtime tradeoffs against lower-level C-native approaches.